Module 1: Matrix Mathematics

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1 Why Matrix Mathematics is Essential for UAV Dynamics and Control

Modern UAV dynamics and control are naturally formulated in matrix and vector form. The UAV state, including position, velocity, orientation, and angular rates, is represented as a vector, while the coupled equations governing translational and rotational motion are compactly expressed using matrices. Coordinate transformations between inertial, body-fixed, and wind frames rely on orthogonal rotation matrices, and inertia properties are encoded in inertia matrices. The stability and performance of UAV control systems are analyzed using the eigenvalues and eigenvectors of system matrices, which reveal dynamic modes such as oscillations, damping, and divergence. Furthermore, state-space models, observers, and optimal controllers (e.g., LQR and Kalman filters) are built entirely on matrix operations. As a result, a strong foundation in matrix mathematics is indispensable for modeling, analyzing, simulating, and controlling UAVs with precision and rigor.

2 Notation

The set of natural numbers is denoted by $\mathbb{N}$. The set of whole numbers is denoted by $\mathbb{W}$. The set of integers is denoted by $\mathbb{Z}$. The set of real numbers is denoted by $\mathbb{R}$. The set of complex numbers is denoted by $\mathbb{C}$. The equation $a \triangleq b$ implies that a is defined as the object b.

Fact 2.1. An element of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{R}$, or $\mathbb{C}$ is a **scalar**.

Remark 1. Scalars satisfy the usual addition and multiplication operations.

Definition 2.1. The **absolute value** of $\alpha \in \mathbb{R}$, denoted by $|\alpha| \geq 0$, is

$$|\alpha| \triangleq \begin{cases} -\alpha, & \alpha < 0, \\ \alpha, & \alpha \geq 0. \end{cases} \tag{1}$$

3 Complex Numbers

Definition 3.1. A **complex number** z is $z = \alpha + \jmath\beta$ where $\alpha, \beta \in \mathbb{R}$ and $\jmath \triangleq \sqrt{-1}$. The set of all complex numbers is denoted by $\mathbb{C}$.

Definition 3.2. The **complex conjugate** of z, denoted by $\overline{z}$, is $\overline{z} = \alpha - \jmath\beta$.

Definition 3.3. The **absolute value, modulus, or magnitude** of z, denoted by $|z| \geq 0$, is

$$|z| = \sqrt{\alpha^2 + \beta^2}. \tag{2}$$

Fact 3.1. Let $z_1, z_2 \in \mathbb{C}$. Then,

$$z_1 + z_2 \triangleq \alpha_1 + \alpha_2 + \jmath(\beta_1 + \beta_2), \tag{3}$$

$$z_1 z_2 \triangleq \alpha_1\alpha_2 - \beta_1\beta_2 + \jmath(\alpha_1\beta_2 + \alpha_2\beta_1), \tag{4}$$

$$\frac{z_1}{z_2} \triangleq \frac{z_1\overline{z}_2}{z_2\overline{z}_2} = \frac{\alpha_1\alpha_2 + \beta_1\beta_2 + \jmath(\alpha_1\beta_1 - \alpha_2\beta_2)}{\alpha_2^2 + \beta_2^2}. \tag{5}$$

3.1 Problems

Problem 3.1. Let $z_1 = 1 + 2\jmath$ and $z_2 = -2 + 1\jmath$. Compute $z_1 + z_2$, $z_1 z_2$, and z_1/z_2.

4 Vectors

Definition 4.1. An n-dimensional **vector** x is a column of n scalars $x_1, x_2, \ldots, x_n$, that is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}. \tag{6}$$

Remark 2. For all $i \in \{1, 2, \ldots, n\}$, if $x_i \in \mathbb{R}$, then x is an n-dimensional real vector. For any $i \in \{1, 2, \ldots, n\}$, if $x_i \in \mathbb{C}$, then x is an n-dimensional complex vector.

Remark 3. A vector is denoted by lowercase letters.

Remark 4. The ith element of x is denoted by x_i.

Definition 4.2. The **dimension** of x is the number of scalar elements in the vector.

Remark 5. The set of n-dimensional real vectors is denoted by $\mathbb{R}^n$. The set of n-dimensional complex vectors is denoted by $\mathbb{C}^n$.

Fact 4.1. Two vectors can be added if and only if they have the same dimension. Let $x, y \in \mathbb{R}^n$. Then,

$$z \triangleq x + y = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}. \tag{7}$$

Fact 4.2. A vector can be multiplied by a scalar, which results in another vector. Let $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}^n$. Then,

$$\alpha x \triangleq \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \tag{8}$$

Fact 4.3. The dot product of two vectors $x, y \in \mathbb{R}^n$, denoted by $x \bullet y \in \mathbb{R}$, is

$$x \bullet y \triangleq \sum_{i=1}^{n} x_i y_i = x_1 y_1 + x_2 y_2 + \ldots + x_n y_n. \tag{9}$$

Note that $x \bullet y = y \bullet x$.

Definition 4.3. The **length, magnitude, or norm** of the vector x, denoted by $\|x\| \in [0, \infty]$, is

$$\|x\| \triangleq \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}. \tag{10}$$

Definition 4.4. Let $x \in \mathbb{R}^n$. If $\|x\| = 1$, then x is called a **unit vector**.

Definition 4.5. The **angle** between two vectors $x, y \in \mathbb{R}^n$, denoted by $\theta_{xy} \in \mathbb{R}$, is

$$\theta_{xy} \triangleq \cos^{-1}\left(\frac{x \bullet y}{\|x\| \|y\|} \right). \tag{11}$$

Note that $\theta_{xy} = \theta_{yx}$.

Fact 4.4. Let $x \in \mathbb{R}^n$. Then, $\theta_{xx} = 0$.

Definition 4.6. Two vectors $x, y \in \mathbb{R}^n$ are **orthogonal** if $\theta_{xy} = \pi/2$.

Example 4.1. Let $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Then, $\theta_{xy} = \pi/2$.

Remark 6. Let $x, y \in \mathbb{R}^n$ be orthogonal. Then, x is orthogonal to y. Similarly, y is orthogonal to x.

Fact 4.5. Let $x, y \in \mathbb{R}^n$ be orthogonal. Then, $x \bullet y = 0$.

Definition 4.7. The **cross-product** of two vectors $x, y \in \mathbb{R}^3$, denoted by $x \times y \in \mathbb{R}^3$, is

$$x \times y \triangleq \begin{bmatrix} x_2y_3 - x_3y_2 \\ x_3y_1 - x_1y_3 \\ x_1y_2 - x_2y_1 \end{bmatrix}. \tag{12}$$

Fact 4.6. For all $x, y \in \mathbb{R}^3$, $(x \times y) \bullet x = (x \times y) \bullet y = 0$.

Fact 4.7. For all $x, y \in \mathbb{R}^3$, $\|x \times y\| = \|x\|\|y\||\sin\theta_{xy}|$.

4.1 Useful Matlab Functions

- `abs`: Absolute value of real or complex number.
- `length`: Can be used to calculate the dimension of a vector.
- `dot`: Dot product of two vectors.
- `norm`: Norm of a vector.
- `acos`: Can be used to calculate $\cos^{-1}$.
- `cross`: Cross product of 3-dimensional vectors.

4.2 Problems

Problem 4.1. Let $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $y = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$.

1. What is the dimension of x and y ?
2. Compute $x + y$ and $x - y$ and draw them on a 2D plane.
3. Compute $x \bullet y$ and $y \bullet x$.
4. Compute the length of x and y.
5. Compute the angle θ_{xy}.
6. Generate a vector u orthogonal to x. Compute θ_{ux}.

Problem 4.2. Consider a unit cube with one corner at the point $(0, 0, 0)$, and the three edges along the $x-$, $y-$, and $z-$axis. Describe all corners of the cube using 3-dimensional vectors.

5 Matrices

Definition 5.1. A $n \times m$ matrix is an array of n rows and m columns of scalars, that is,

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix}. \tag{13}$$

The matrix A above has n rows and m columns.

Definition 5.2. The dimension of a matrix is the number of rows times the number of columns. The dimension of A above is $n \times m$. If $n > m$, then A is a tall matrix. If $n = m$, then A is a square matrix. If $n < m$, then A is a wide matrix.

Example 5.1. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}. \tag{14}$$

Then, A is a **wide** matrix, B is a **square** matrix, and C is a **tall** matrix.

Remark 7. The set of $n \times m$ real matrices is denoted by $\mathbb{R}^{n\times m}$. The set of $n \times m$ complex matrices is denoted by $\mathbb{C}^{n\times m}$.

Remark 8. A matrix is denoted by uppercase letters.

Remark 9. The element in the ith row and the $j$the column of A is denoted by a_{ij}. The element a_{ij} is also denoted by the (i, j)th element of A and $A(i, j)$.

Fact 5.1. Two matrices can be added if and only if they have the same dimension. Let $A, B \in \mathbb{R}^{n\times m}$. Then,

$$C \triangleq A + B = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1m} + b_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nm} + b_{nm} \end{bmatrix}. \tag{15}$$

Fact 5.2. A matrix can be multiplied by a scalar. Let $\alpha \in \mathbb{R}$ and $A \in \mathbb{R}^{n\times m}$. Then,

$$\alpha A \triangleq \begin{bmatrix} \alpha a_{11} & \cdots & \alpha a_{1m} \\ \vdots & \ddots & \vdots \\ \alpha a_{n1} & \cdots & \alpha a_{nm} \end{bmatrix}. \tag{16}$$

Fact 5.3. Two matrices can be multiplied if and only if the number of columns of the first matrix equals the number of rows of the second matrix. Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{p\times q}$ be two matrices. Then, the matrix product AB is defined if and only if $m = p$.

Fact 5.4. Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times q}$. Define $C \triangleq AB \in \mathbb{R}^{n\times q}$. Then,

$$c_{ij} \triangleq \sum_{k=1}^{m} a_{ik} b_{kj}. \tag{17}$$

Remark 10. Let $A, B \in \mathbb{R}^{n\times n}$. In general, $AB \neq BA$.

Example 5.2. Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \\ -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}. \tag{18}$$

Then,

$$AB = \begin{bmatrix} -2 & -1 & 0 \\ 9 & 12 & 15 \\ 2 & 1 & 0 \end{bmatrix}. \tag{19}$$

Definition 5.3. The $n \times n$ identity matrix I_n is

$$I_n(i,j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{20}$$

Example 5.3. The 2×2 identity matrix I_2 is

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{21}$$

Fact 5.5. Let $A \in \mathbb{R}^{n \times n}$. The $n \times n$ identity matrix I_n satisfies $AI_n = I_n A = A$.

Fact 5.6. A matrix and a vector can be multiplied if and only if the number of columns of the matrix equals the dimension of the vector. Let $A \in \mathbb{R}^{n \times m}$ and $x \in \mathbb{R}^p$. Then, Ax is defined if and only if $m = p$.

Fact 5.7. Let $A \in \mathbb{R}^{n \times m}$ and $x \in \mathbb{R}^m$. Define $y \triangleq Ax \in \mathbb{R}^n$. Then,

$$y_i \triangleq \sum_{k=1}^{m} a_{ik} x_k. \tag{22}$$

Fact 5.8. Let $x \in \mathbb{R}^n$. The $n \times n$ identity matrix I_n satisfies $I_n x = x$.

Definition 5.4. Let $A \in \mathbb{R}^{n \times n}$. Then,

$$A^p \triangleq \underbrace{AA \cdots A}_{p \text{ times}}. \tag{23}$$

Definition 5.5. Let $A \in \mathbb{R}^{n \times m}$. Then, the transpose of A is an $m \times n$ matrix such that the (i,j)th element of the transpose of A is a_{ji}. The transpose of A is denoted by A^{T}. Note that $A^{\mathrm{T}} \in \mathbb{R}^{m \times n}$.

Example 5.4. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}. \tag{24}$$

Then,

$$A^{\mathrm{T}} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}. \tag{25}$$

Fact 5.9. Let $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times p}$. Then, $(AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}} \in \mathbb{R}^{p \times n}$.

Definition 5.6. Let $x \in \mathbb{R}^n$. Then, $x^{\mathrm{T}} \in \mathbb{R}^{1 \times n}$ is called a **row vector**.

Fact 5.10. Let $x, y \in \mathbb{R}^n$. Then,

$$x \bullet y = x^{\mathrm{T}} y. \tag{26}$$

Fact 5.11. Let $x, y \in \mathbb{R}^n$. Then, $x^{\mathrm{T}} y \in \mathbb{R}$ and $xy^{\mathrm{T}} \in \mathbb{R}^{n \times n}$.

Definition 5.7. The **trace** of $A \in \mathbb{R}^{n \times n}$, denoted by $\operatorname{tr} A$, is the sum of its diagonal elements.

$$\operatorname{tr} A \triangleq \sum_{i=1}^{n} a_{ii}. \tag{27}$$

Definition 5.8. The **determinant** of $A \in \mathbb{R}^{n \times n}$, denoted by $\det(A) \in \mathbb{R}$, is given by the recursive expression

$$\det(A) \triangleq (-1)^{i-1} \sum_{j=1}^{n} (-1)^{j-1} a_{ij} \det(M_{ij}), \tag{28}$$

where $M_{ij} \in \mathbb{R}^{(n-1) \times (n-1)}$ is the $(n-1) \times (n-1)$ matrix formed by removing the ith row and jth column of A. For simplicity, we choose $i = 1$.

Fact 5.12. Let $A \in \mathbb{R}^{1\times 1}$. The determinant of A is given by $\det(A) = A$.

Fact 5.13. Let $A \in \mathbb{R}^{2\times 2}$. The determinant of A is given by

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}. \tag{29}$$

Fact 5.14. Let $A \in \mathbb{R}^{3\times 3}$. The determinant of A is given by

$$\begin{aligned}\det(A) &= a_{11}\det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{bmatrix}\right) - a_{12}\det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{bmatrix}\right) + a_{13}\det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{bmatrix}\right) \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).\end{aligned} \tag{30}$$

Definition 5.9. Let $A, B \in \mathbb{R}^{n\times n}$ be two square matrices. Then, B is an **inverse** of A if and only if

$$AB = BA = I_n. \tag{31}$$

The inverse of A, if it exists, is denoted by A^{-1}.

Remark 11. All square matrices are not invertible.

Remark 12. In general, computation of the inverse of a matrix is a difficult process.

Fact 5.15. The square matrix A is invertible if and only if $\det(A) \neq 0$.

Definition 5.10. Let $A \in \mathbb{R}^{n\times n}$ and suppose that A^p is invertible. Then, $A^{-p} \triangleq (A^p)^{-1}$.

Fact 5.16. Let $A \in \mathbb{R}^{2\times 2}$. In the case where $\det(A) = a_{11}a_{22} - a_{12}a_{21} \neq 0$, the inverse of A is given by

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11}\end{bmatrix}. \tag{32}$$

Fact 5.17. Let $x, y \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ such that $y = Ax$. If A is invertible, then $x = A^{-1}y$.

Definition 5.11. Let $A \in \mathbb{R}^{n\times n}$. If each column of A is unit-length and orthogonal to every other column of A, then A is an **orthogonal** matrix.

Fact 5.18. Let $A \in \mathbb{R}^{n\times n}$ be orthogonal. Then, $A^{-1} = A^{\mathrm{T}}$. Consequently, $AA^{\mathrm{T}} = A^{\mathrm{T}}A = I_n$.

Remark 13. Orthogonal matrices preserve vector length and represent rigid body rotations.

Example 5.5. Let

$$R = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{bmatrix}. \tag{33}$$

Then, R is an orthogonal matrix. Note that

$$R^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix} = R^{\mathrm{T}}. \tag{34}$$

Definition 5.12. Let $A \in \mathbb{R}^{n\times n}$. Then, $v \in \mathbb{C}^n$ is an eigenvector of A and $\lambda \in \mathbb{C}$ is an eigenvalue of A if

$$Av = \lambda v. \tag{35}$$

Fact 5.19. Let λ be an eigenvalue and v be an eigenvector of $A \in \mathbb{R}^{n\times n}$. Then,

$$\det(\lambda I_n - A) = 0. \tag{36}$$

5.1 Useful Matlab Functions

- `size`: Can be used to calculate the dimension of a matrix.
- `eye`: Creates identity matrix of chosen dimensions.
- `transpose`: Transpose of a matrix.
- `trace`: Trace of a square matrix.
- `det`: Determinant of a square matrix.
- `inv`: Inverse of an invertible matrix.
- `eig`: Obtains eigenvalues and eigenvectors of a matrix.
- `ones`: Creates a matrix whose elements are all 1.
- `zeros`: Creates a matrix whose elements are all 0.

5.2 Problems

Problem 5.1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix}$.

1. Compute $A + B$ and $A - B$.
2. Compute AB and BA. Are they equal?
3. Compute $A + A^{\mathrm{T}}$ and $B + B^{\mathrm{T}}$.
4. Compute $\det(A)$ and $\det(B)$.
5. Are the matrices A and B orthogonal?
6. Using Matlab, compute A^{-1} and B^{-1}. Verify that $AA^{-1} = BB^{-1} = I_2$.